



Take Home Challenge: Senior Infrastructure Engineer

We've designed the following task for you. Impress us with your ideas and your obsession for simplicity.

The Challenge

As a member of the Prisma Infrastructure team, you will work on the core infrastructure our team and customers depend on. You'll work across tech stacks to secure our products, design distributed systems, and handle increasing scale. To get an idea of your experience and your approach, we'd like you to complete the task below and answer the follow-up questions related to it.

The Task

Your task is to build and automate a system for an application development team that deploys web server containers dynamically. Using Infrastructure as Code (IaC) and configuration management, scaffold a fresh Debian host where an internally-facing service can create and remove identical containerised web-server instances, and a reverse proxy exposes them externally through HTTP/S. A

single host deployment is acceptable. Apply system hardening best practices, enforce container security basics, demonstrate secure networking choices, and keep it reproducible end-to-end.

We expect this assignment to take no more than **4 hours**, and should be **submitted within 5 days** of its receipt. This is a broad task with a short time and we favour completed individual features over completing all requirements. After submitting the assessment, a technical interview will be scheduled with the Prisma team to discuss your approach, decisions, and trade-offs. While not a line-by-line code review, you should be prepared to discuss specific sections of code in this session.

Apart from the requirements below, you're welcome to use any tools to accomplish the task. Feel free to use any AI tooling that is part of your normal workflow. Please annotate which parts of the codebase were generated and be prepared to discuss these choices in the technical interview. Just ensure the submission is your own, avoids excessive auto-generated code, and shows clear reasoning behind your decisions.

System Requirements

1. A single Debian host (single-node acceptable) that can be recreated end-to-end using Infrastructure as Code and configuration management (tools are your choice). The system must let an internal interface create and remove identical containerised web-server instances. Each web server instance should be externally accessible via HTTP/S.
2. Apply pragmatic Linux hardening.
3. Enforce basic container security.
4. External traffic must be terminated or routed correctly.

Technical Requirements

1. Reproducible provisioning and configuration of the host, network, and runtime (use any IaC/CM tools you prefer).
2. A minimal internal control surface (API or CLI) to create and delete instances, register routing, and list active instances.

3. Route external requests to the correct instance.
4. Basic verification: a single command/script that asserts expected open ports and HTTP 200 for a created web server.

Out-of-scope

- Building custom container images per instance (all instances may use the same image).
- Full multi-host orchestration, global routing, or production-grade HA.
- Complex CI/CD, vulnerability scanning, or secret management backends; keep secrets as templates/mocks only.
- Vendor lock-in to specific clouds or tools, choose what you work fastest with.

Deliverables



Important: For us, the how and the why are important. Sending us your results somewhat incomplete is totally fine.

- **Codebase:** Provide the code in a public or private GitHub repository. If you prefer a private repo, please invite the following people to your private repo: `Nova-ZA` ; `rtbenfield` ; `ellotheth` ; `tylerhogarth` .
- **Documentation:** A README that covers setup, how to provision an instance, how to run verification, and a short rationale for your security and networking choices.

Submission

Please submit your assignment in the form of the Github link via Rippling. The link is provided in the email.

If you run into any problems, please reach out to us, we are happy to help. We wish you all the best with the assignment, and look forward to see what you come up with!
